Winning Space Race with Data Science

<Name>
<Date>

Khalid Mohammed | Oct 16, 2026 | GitHub: https://github.com/KhalidMohd99/Data-Science-Capstone-Project



Outline

- 1. Executive Summary
- 2. Introduction
- 3. Methodology
- 4. Results
- 5. Conclusions
- 6. Appendix

Executive Summary

- Project: Predict Falcon 9 first-stage landing success (Class 1=success, 0=failure).
- Data: 101 launches (SpaceX REST API + Wikipedia scraping); 90 records modelled after wrangling.
- EDA: 6 visualizations + 10 SQL queries (launch sites, payload ranges, orbits, temporal trends).
- Interactive: Folium launch site map + Plotly Dash interactive dashboard.
- ML: Logistic Regression, SVM, Decision Tree, KNN; best model = Logistic Regression (test accuracy 83.3%).

Introduction

- Background: SpaceX advertises Falcon 9 launches at \$62M; reusability of the first stage is critical to lowering costs.
- Problem: Can we predict whether the first stage of a Falcon 9 launch will land successfully?
- Sub-questions: Which factors (launch site, orbit, payload mass, booster version) best predict success?
- Business value: Accurate prediction helps estimate launch costs, plan missions, and assess risk.
- Data: Public SpaceX launch data via REST API (v4) and Wikipedia table scraping.
- Approach: EDA (vis + SQL), interactive analytics (Folium + Dash), supervised classification (4 models).
- Complete work available at:

Section 1

Methodology



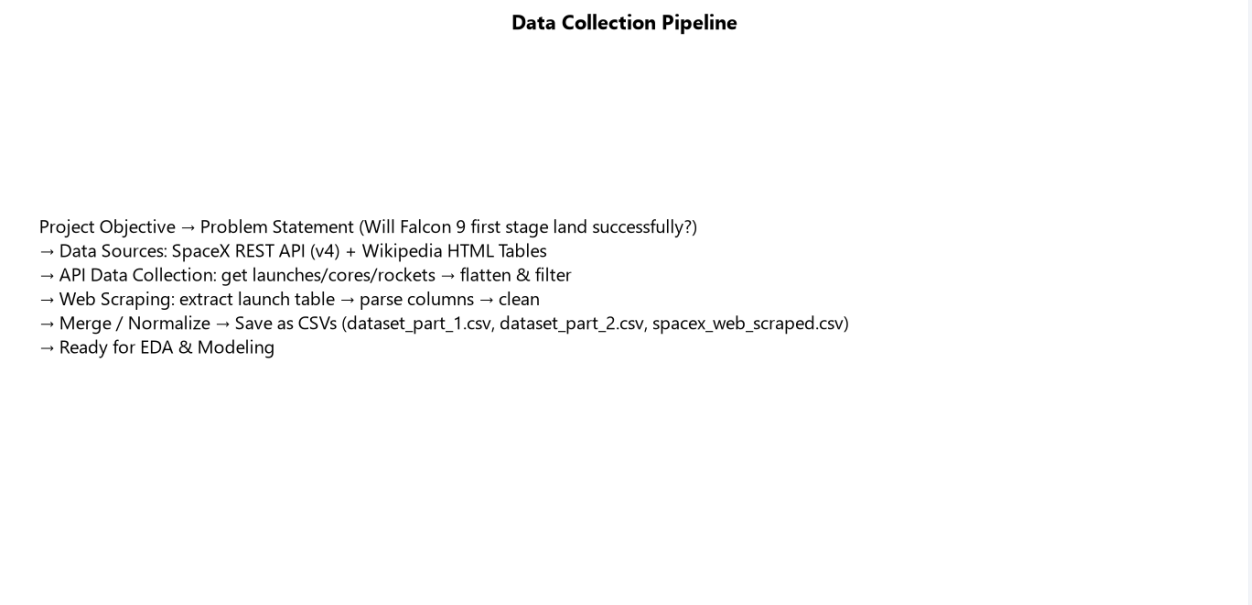
Methodology Overview

Data Collection: SpaceX API v4 + Wikipedia HTML scraping (BeautifulSoup).

- Data Wrangling: Filtering, cleaning, date parsing, feature extraction, one-hot encoding.
- EDA (Visualization): 6 charts (site vs flight number/payload, orbit analysis, yearly trend).
- EDA (SQL): 10 targeted queries on SPACEXTBL/SPACEXTABLE (location, payload sums/averages, outcomes).
- Interactive Analytics: Folium map (markers/clusters, distances to coastline/city/highway).
- Interactive Dashboard: Plotly Dash with dropdown, range slider, pie & scatter charts.
- Predictive Modeling: LR, SVM, Decision Tree, KNN; 80/20 split; GridSearchCV 10-fold; model comparison & selection.
- GitHub: https://github.com/KhalidMohd99/Data-Science-Capstone-Project

Data Collection Methodology

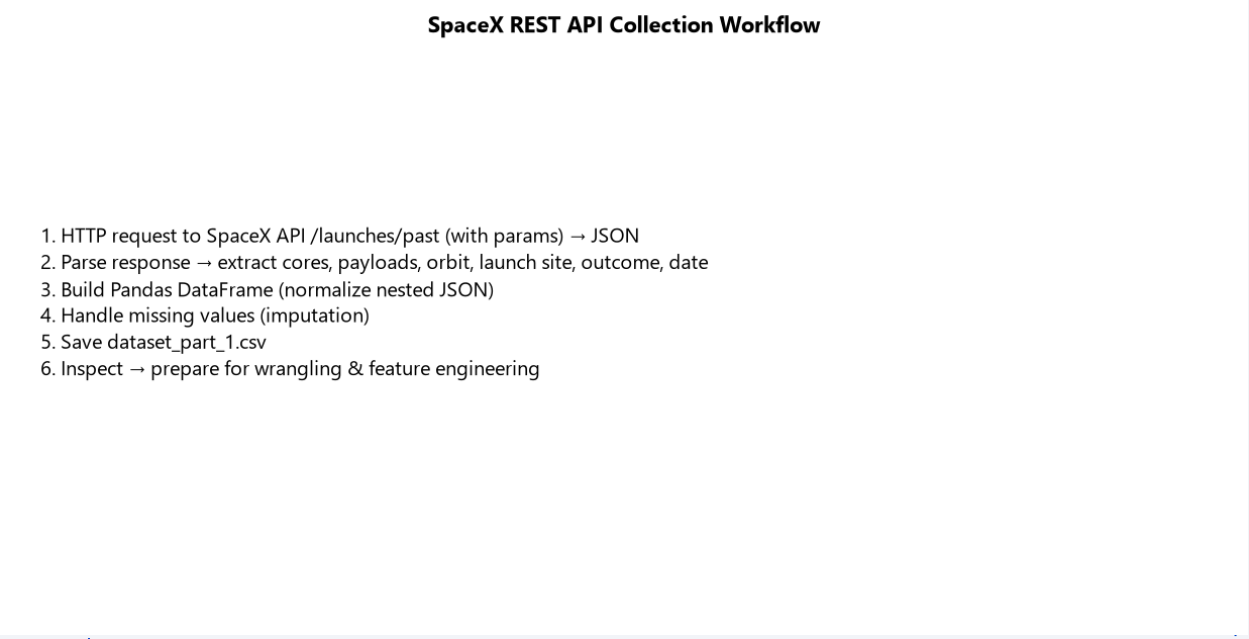
- Sources: SpaceX REST API (v4) — launches, rockets, cc
- API calls: retrieve JSON; extract fields (launch site, pay
- Web scraping: requests + BeautifulSoup → locate HTM
- Output: dataset_part_1.csv, spacex_web_scraped.csv
- Flow: Data Sources → API extraction → Scraping → Cl
- Completed notebooks: 01-API + 02-Scraping (see http



Data Collection – SpaceX REST API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts

- Add the GitHub URL of the completed SpaceX API calls notebook (must include completed code cell and outcome cell), as an external reference and peer-review purpose



Data Collection – Web Scraping (Wikipedia)

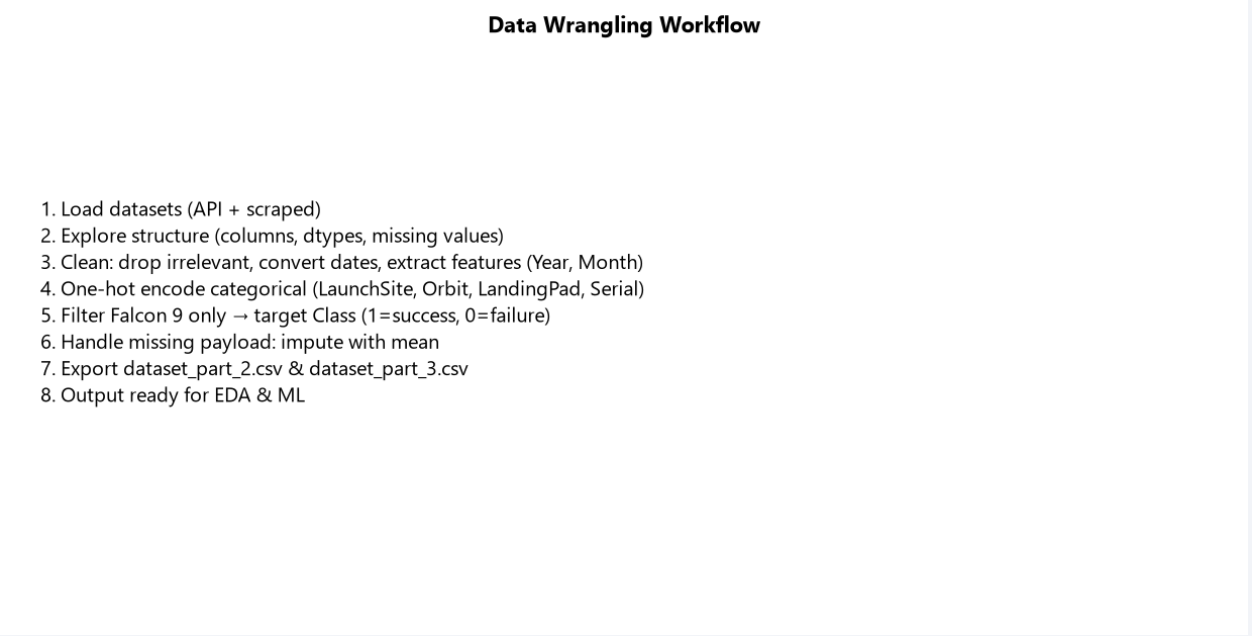
Data Collection - Scraping

- Goal: Complement API with missing details from Wikipedia table.
- Tools: requests + BeautifulSoup + regular expressions; custom extract_column_from_header.
- Steps: fetch HTML → extract wikitable → parse headers/rows → cast types → export CSV.
- Performed in: 02-jupyter-labs-webscraping.ipynb (fixed & executed; outputs saved).
- GitHub: https://github.com/KhalidMohd99/Data-Science-Capstone-Project/blob/main/02-jupyter-labs-webscraping.ipynb



Data Wrangling Methodology

- Objectives: Clean, consolidate and engineer features f
- Tasks: handle missing values (PayloadMass mean), filt
- Encoding: one-hot encode LaunchSite, Orbit, Landing
- Target: Class (1=success, 0=failure). Records after wra
- Performed in: 03-labs-jupyter-spacex-Data wrangling.
- GitHub: https://github.com/KhalidMohd99/Data-Scie spacex-Data%20wrangling.ipynb



EDA with Data Visualization – Methodology

- Approach: Explore relationships between flight number, launch site, payload mass, orbit and success rate.
- Charts produced (6):
- 1. Flight Number vs Launch Site (scatter/catplot)
- 2. Payload Mass vs Launch Site (scatter/catplot)
- 3. Success Rate vs Orbit Type (bar)
- 4. Flight Number vs Orbit Type (scatter/catplot)
- 5. Payload Mass vs Orbit Type (scatter/catplot)
- 6. Launch Success Yearly Trend (line)
- Tools: pandas, matplotlib, seaborn. Performed in jupyter-labs-eda-dataviz-v2.ipynb.
- GitHub: https://github.com/KhalidMohd99/Data-Science-Capstone-Project/blob/main/jupyter-labs-eda-dataviz-v2.ipynb

EDA with SQL – Methodology

- Database: SQLite (04-my_data1.db) with table SPACEXTBL/SPACEXTABLE.
- Queries (10): unique sites, CCA-starting launches, total payload (NASA CRS), average payload (F9 v1.1), first ground-pad success date, drone-ship successes 4000-6000kg, mission outcome totals, max-payload booster versions (subquery), 2015 drone-ship failures, rank outcomes 2010-06-04 to 2017-03-20.
- Executed via ipython-sql in 04-SQL Notebook for Peer .ipynb.
- GitHub: https://github.com/KhalidMohd99/Data-Science-Capstone-Project/blob/main/04-SQL%20Notebook%20for%20Peer%20.ipynb

Interactive Map with Folium – Methodology

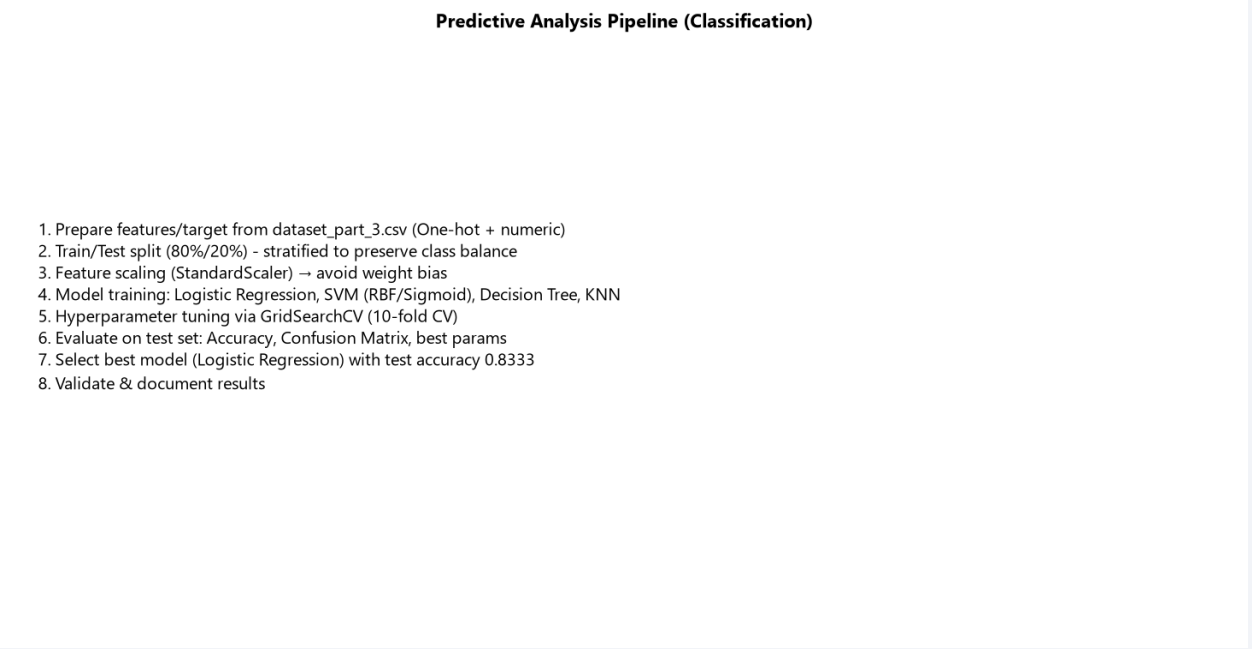
- Purpose: Visualize launch site locations, outcomes and proximity (coastline/city/highway).
- Objects used: folium.Map, folium.Circle, folium.Marker, DivIcon, MarkerCluster, PolyLine, distance calculation.
- Analysis: proximity of CCAFS SLC-40 to coastline (0.58 km), Port Canaveral (17.2 km), I-95 (19.8 km).
- Implemented in launch_site_location.ipynb (fixed & executed).
- GitHub: https://github.com/KhalidMohd99/Data-Science-Capstone-Project/blob/main/launch_site_location.ipynb

Build Dashboard with Plotly Dash – Methodology

- Components: Dropdown (launch site selector: All Sites / specific site), RangeSlider (payload mass filter).
- Graphs: Pie chart (total successful launches per site; success vs failure for selected site), Scatter (Payload vs Launch Outcome with hover info).
- Callbacks: responsive updates on dropdown + slider changes.
- App file: spacex_dash_app.py (runnable). Notebook: 06-plotly-dash-dashboard.ipynb.
- GitHub: https://github.com/KhalidMohd99/Data-Science-Capstone-Project/blob/main/spacex_dash_app.py

Predictive Analysis (Classification) – Methodology

- Models: Logistic Regression, SVM (with kernel tuning)
- Preprocessing: one-hot features (dataset_part_3.csv),
- Tuning: GridSearchCV with 10-fold cross-validation to
- Metrics: Accuracy (test) + Confusion Matrix. Best moc
- Notebook: Modul-04_SpaceX_Machine Learning Pred
- GitHub: https://github.com/KhalidMohd99/Data-Scie
  04_SpaceX_Machine%20Learning%20Prediction_Part



Results

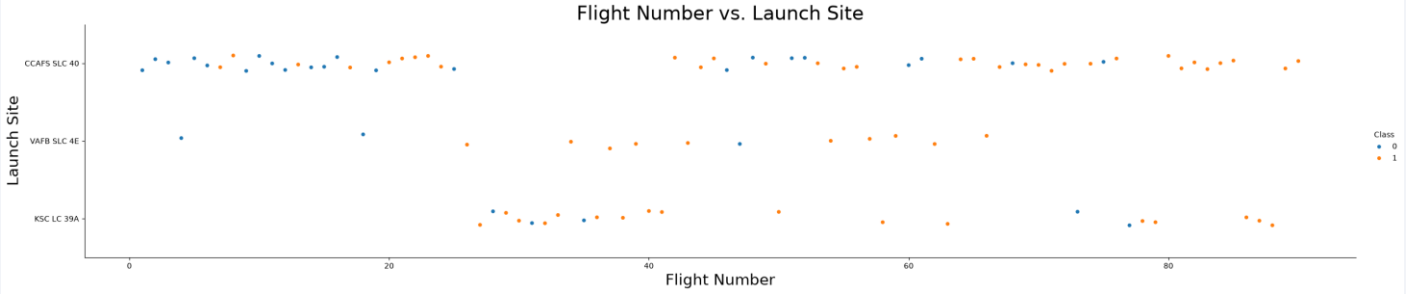
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Section 2

Insights drawn from EDA

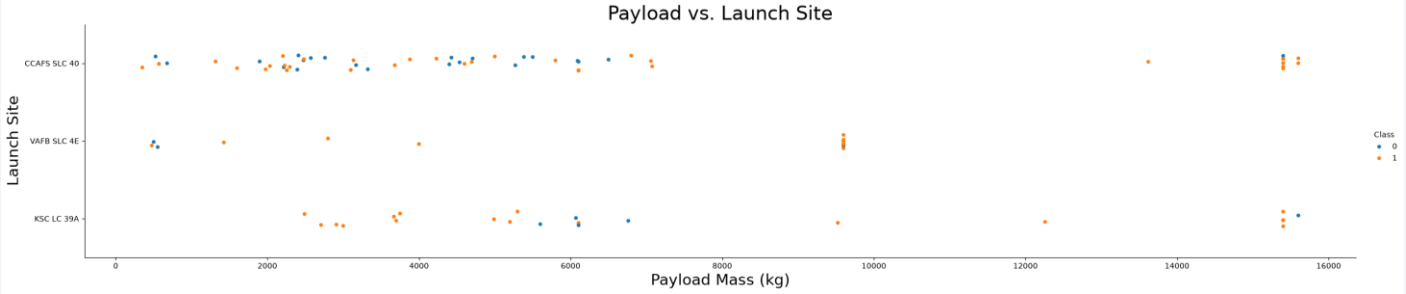
EDA Results – Flight Number vs Launch Site

- Scatter/catplot showing flight progression by launch site and landing outcome (Class). Early flights: higher failure rate; later flights: higher success rate at KSC LC-39A.



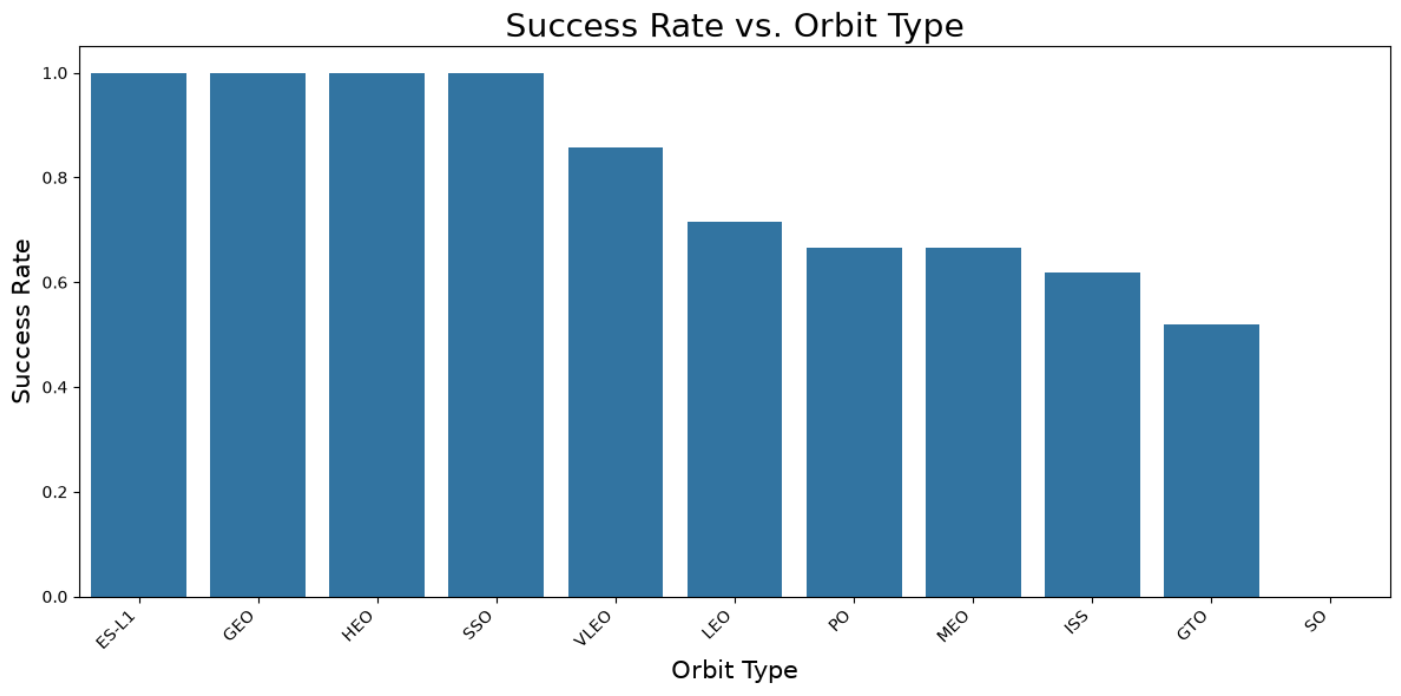
EDA Results – Payload Mass vs Launch Site

- Payload mass distribution across sites. Higher payloads (>10t) at KSC/CCAFS show stronger success association.



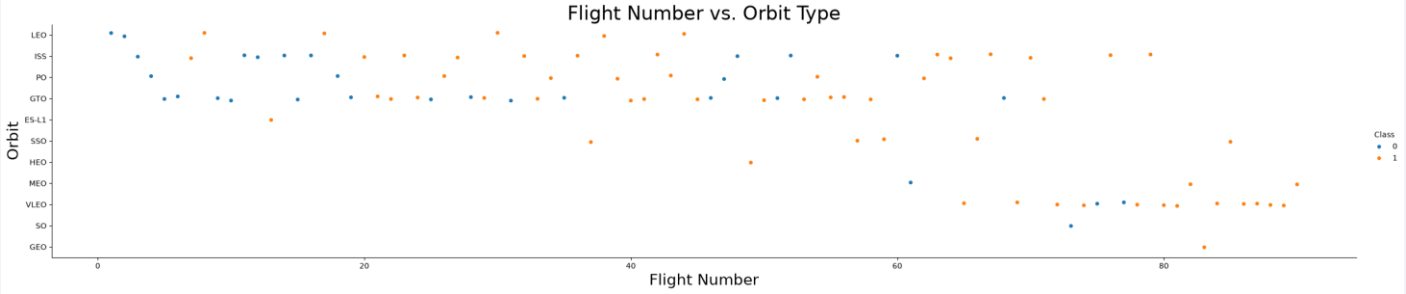
EDA Results – Success Rate by Orbit Type

- Orbits ES-L1/GEO/HEO/SSO/VLEO exhibit high success rates; GTO (27 launches) is most challenging with ~51.85% success.



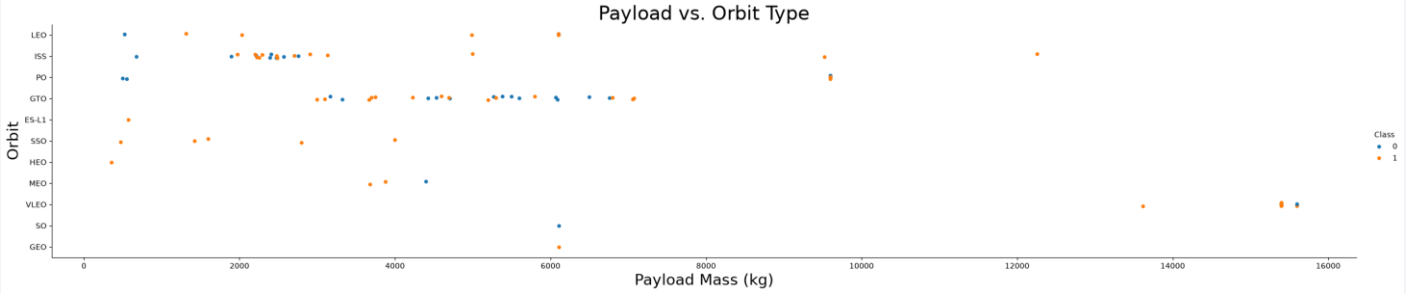
EDA Results – Flight Number vs Orbit Type

- Flight number progression reveals learning curve: later flights increasingly target high-value orbits (VLEO/LEO) with improved success rates.



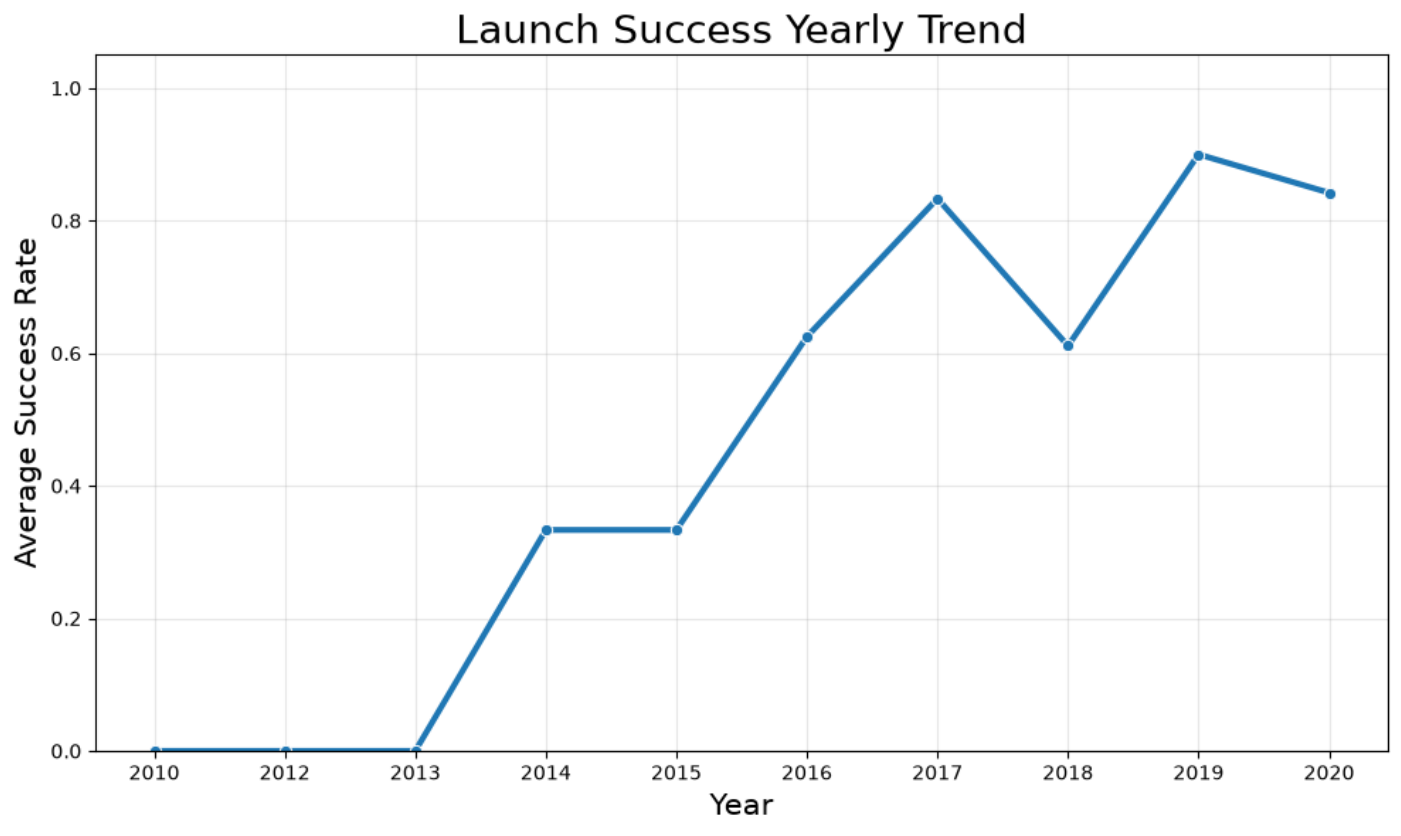
EDA Results – Payload Mass vs Orbit Type

- Orbit-specific payload patterns: LEO/ISS support moderate masses; GTO supports heavy payloads with lower success rate.



EDA Results – Launch Success Yearly Trend

- Success rate improved markedly over time: 0% (2010-2013) → rising to ~90% in 2019 → ~84% in 2020, reflecting engineering/process improvements and reusability maturity.



SQL 1 – Unique Launch Site Names

- Distinct launch sites: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E.

SQL Result 1

```
SELECT DISTINCT Launch_Site
FROM SPACEXTABLE;
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

SQL 2 – Launch Sites Starting with 'CCA' (5 records)

- Sample records from CCAFS show early launch history and outcome patterns.

SQL Result 2

```
SELECT * FROM SPACEXTABLE
WHERE Launch_Site LIKE 'CCA%'
LIMIT 5;
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

SQL 3 – Total Payload Mass (NASA CRS)

- Total payload carried by NASA CRS missions: SUM(PAYLOAD_MASS__KG_) from SPACEXTABLE.

SQL Result 3

```
SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass
FROM SPACEXTABLE
WHERE Customer = 'NASA (CRS)';
```

Total_Payload_Mass
45596

SQL 4 – Average Payload Mass (Booster F9 v1.1)

- Average payload for booster version F9 v1.1 and variants: computed via AVG.

SQL Result 4

```
SELECT AVG(PAYLOAD_MASS__KG_) AS Average_Payload_Mass
FROM SPACEXTABLE
WHERE Booster_Version LIKE 'F9 v1.1%';
```

Average_Payload_Mass
2534.666666666665

SQL 5 – First Successful Ground-Pad Landing Date

- First successful ground-pad landing date identified using MIN(Date).

SQL Result 5

```
SELECT MIN(Date) AS First_Success_Ground_Pad
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (ground pad)';
```

First_Success_Ground_Pad
2015-12-22

SQL 6 – Drone-Ship Successes with Payload 4000–6000 kg

- Boosters achieving drone-ship success in the 4000–6000 kg band.

SQL Result 6

```
SELECT Booster_Version
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)'
  AND PAYLOAD_MASS__KG_ > 4000
  AND PAYLOAD_MASS__KG_ < 6000;
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

SQL 7 – Mission Outcome Totals (Success/Failure)

- Counts grouped by Mission_Outcome (Success/Failure/Partial/Prelaunch Failure) highlight mission reliability.

SQL Result 7

```
SELECT Mission_Outcome, COUNT(*) AS Total
FROM SPACEXTABLE
GROUP BY Mission_Outcome;
```

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

SQL 8 – Booster Versions Carrying Maximum Payload (Subquery)

- Booster versions that carried the maximum payload mass (identified via subquery MAX).

SQL Result 8

```
SELECT DISTINCT Booster_Version
FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

SQL 9 – 2015 Drone-Ship Failures (Month/Version/Site/Outcome)

- Month, outcome, booster version and launch site for 2015 drone-ship failures.

SQL Result 9

```
SELECT substr(Date, 6, 2) AS Month, Landing_Outcome, Booster_Version, Launch_Site
FROM SPACEXTABLE
WHERE substr(Date, 1, 4) = '2015'
  AND Landing_Outcome = 'Failure (drone ship)';
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

SQL 10 – Ranked Landing Outcomes (2010-06-04 to 2017-03-20)

- Landing outcomes ranked by count (descending) between 2010-06-04 and 2017-03-20.

SQL Result 10

```
SELECT Landing_Outcome, COUNT(*) AS Outcome_Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Outcome_Count DESC;
```

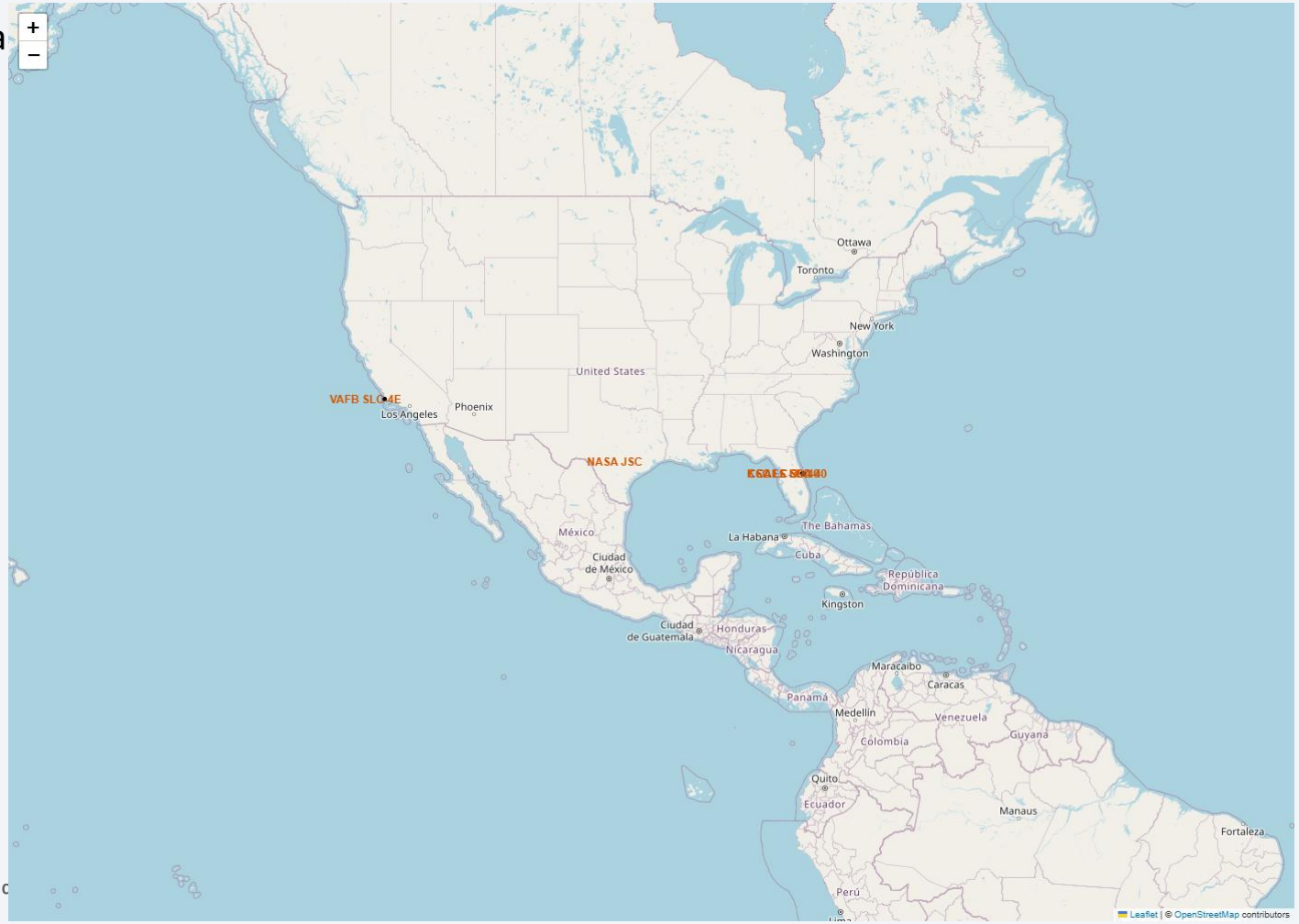
Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

Launch Sites Proximities Analysis

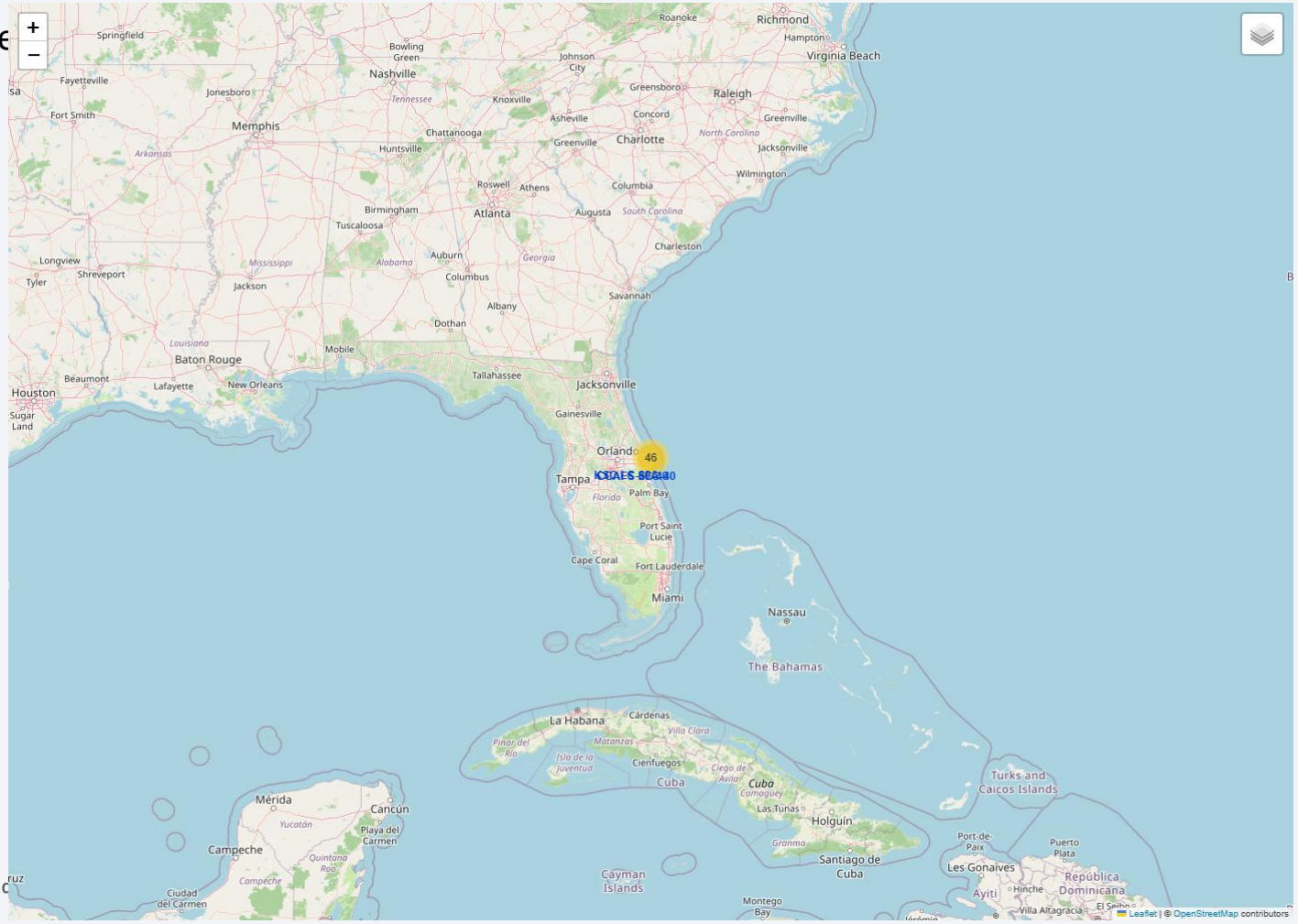
Interactive Map – Launch Sites (Global View)

- All launch sites marked on a global Folium ma
  4E.



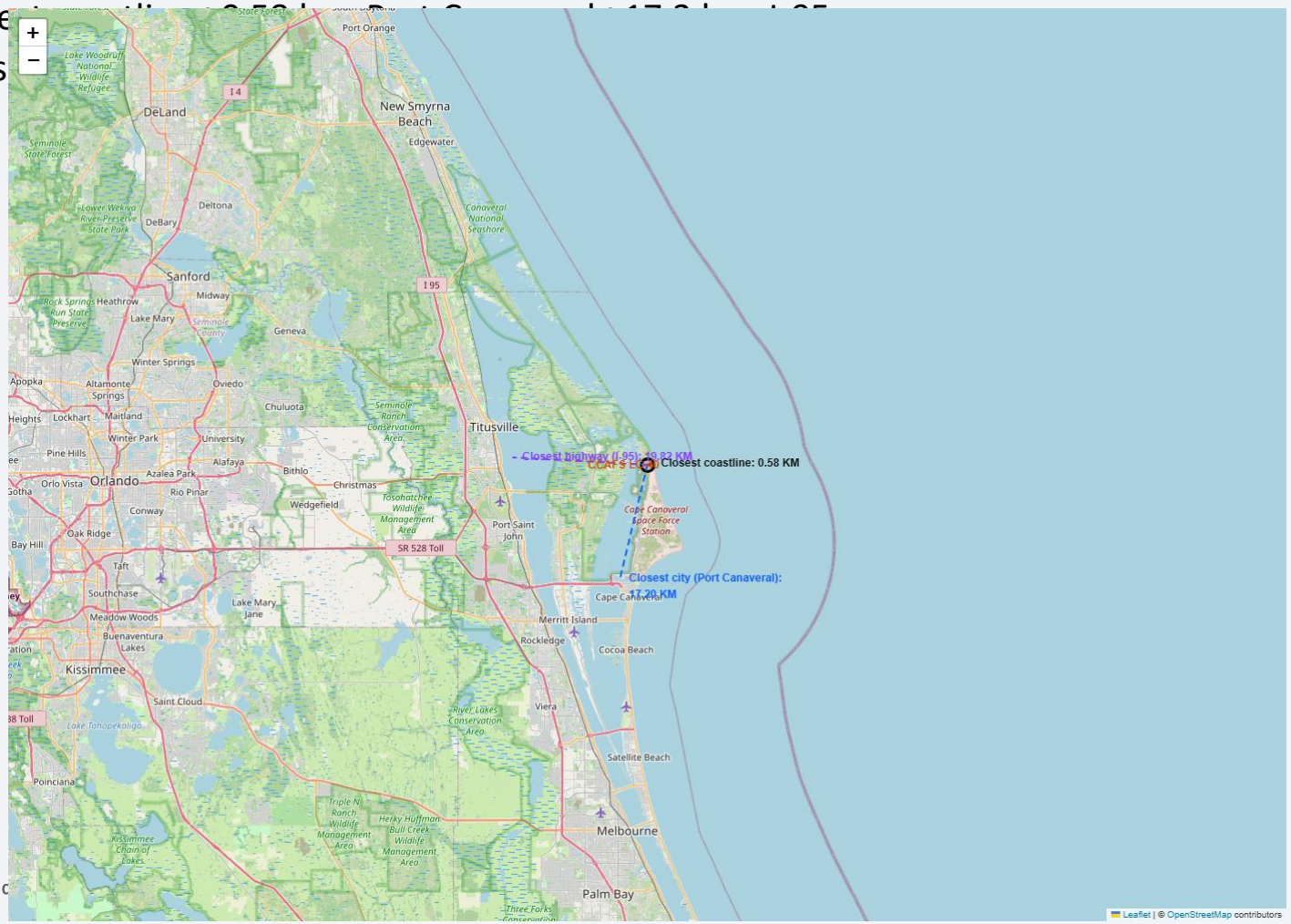
Interactive Map – Launch Outcomes (Color-Coded)

- MarkerCluster with color-coded markers: gree
  reveals site-specific outcome patterns.



Interactive Map – Site Proximity Analysis (CCAFS SLC-40)

- Distances calculated from CCAFS SLC-40: close
  highway ~19.8 km. Proximity to coast enables

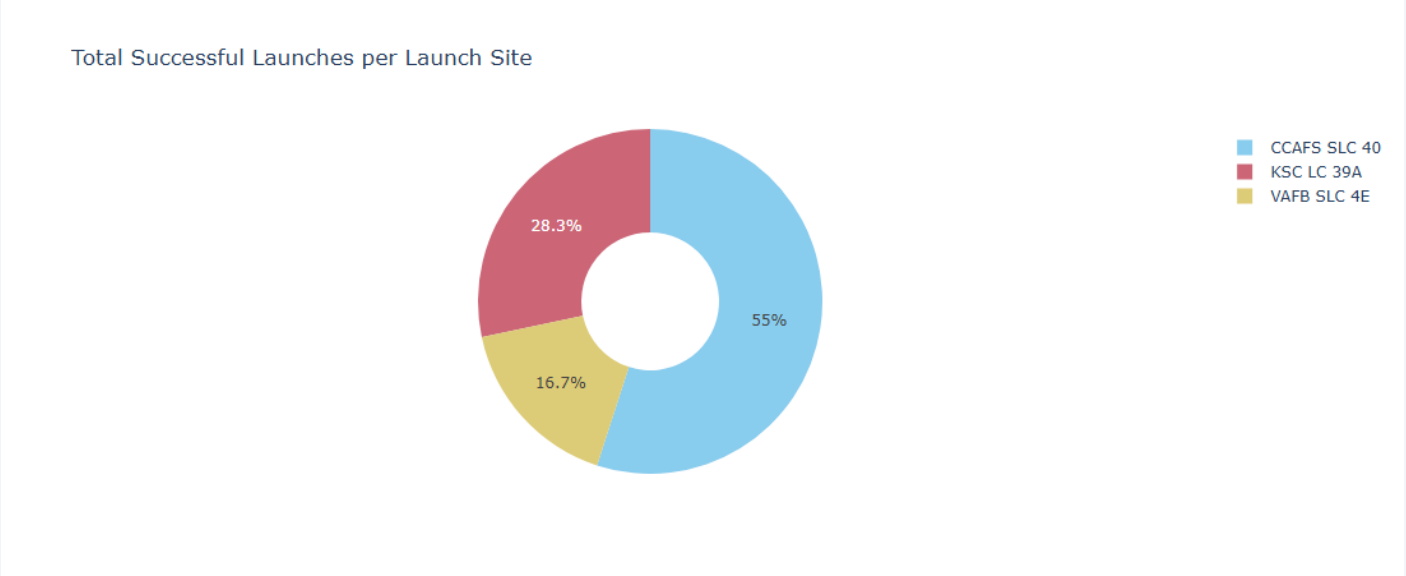


Section 4

Build a Dashboard with Plotly Dash

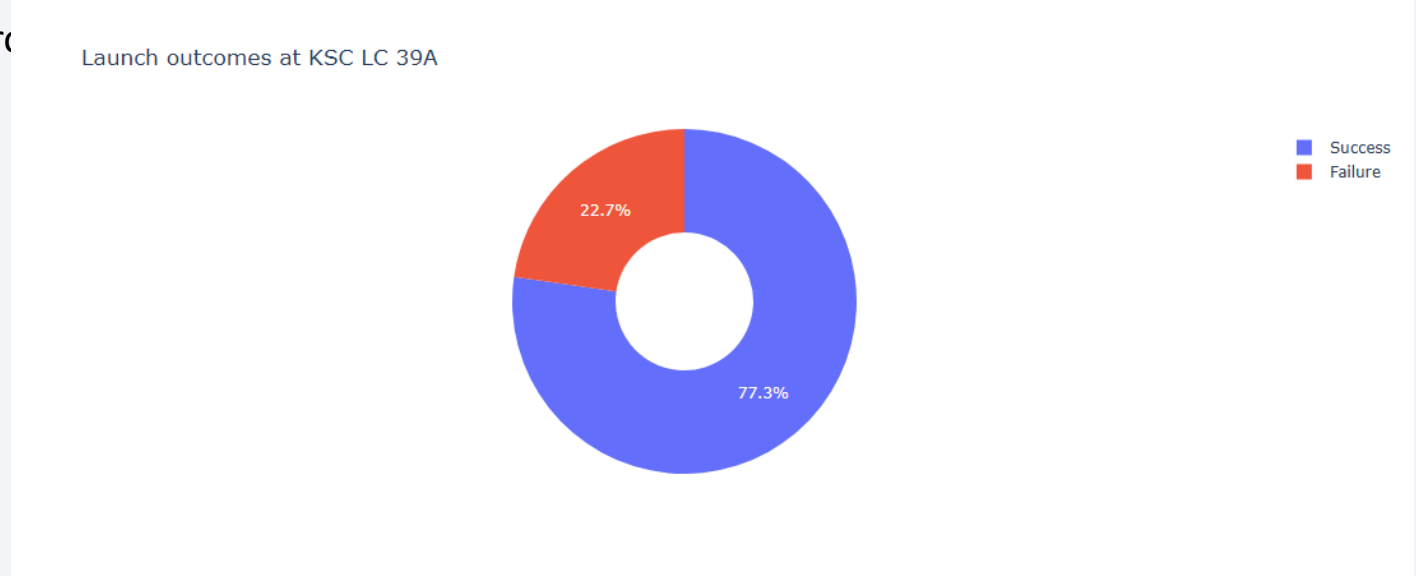
Dashboard – Success Counts per Launch Site (All Sites)

- Plotly Dash pie chart showing total successful (77.3%).



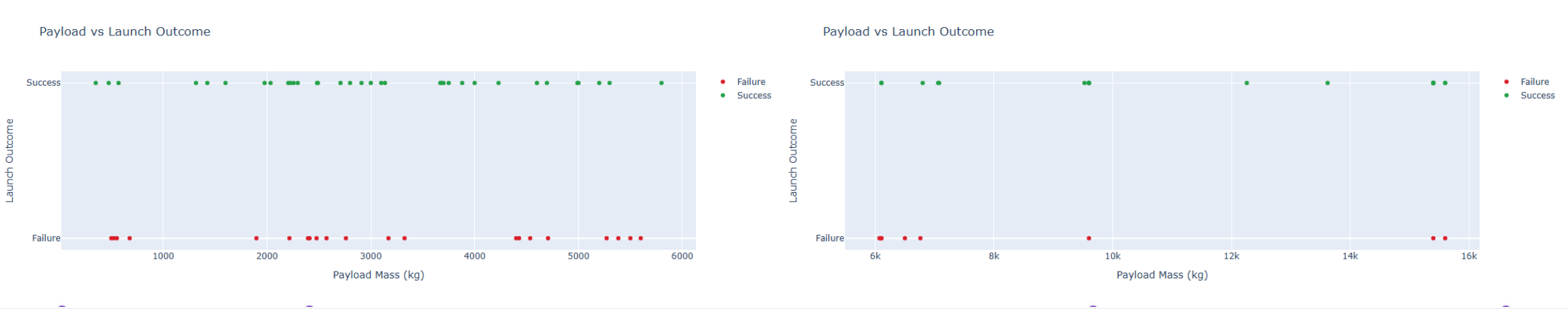
Dashboard – Success vs Failure for Selected Site (KSC LC-39A)

- Interactive pie chart filtered by launch site (dr



Dashboard – Payload vs Launch Outcome (Range Slider)

- Scatter chart: Payload Mass (kg) vs Launch Outcome (0/Failure, 1/Success), color-coded, hover showing flight details. Range slider filters payload bands. Payloads in 0–6000 kg show 61.1% success; 6000–16000 kg show 51.4% success in the filtered view.



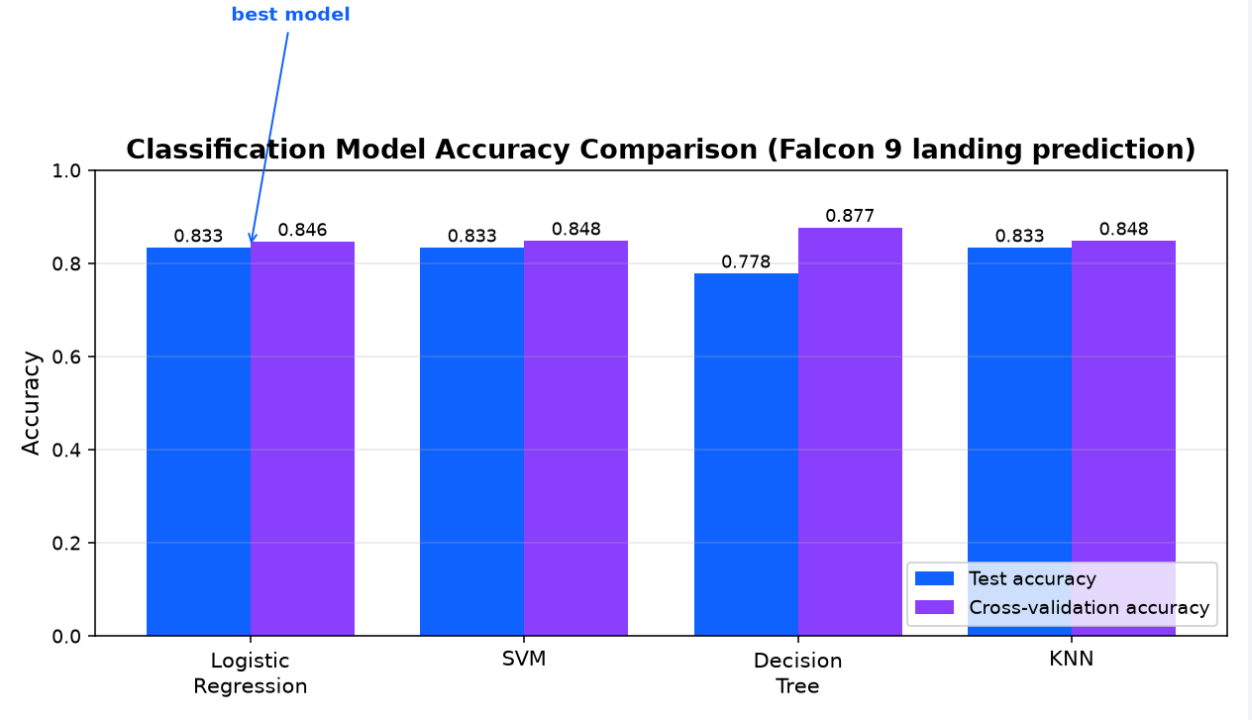
Section 5

Predictive Analysis (Classification)



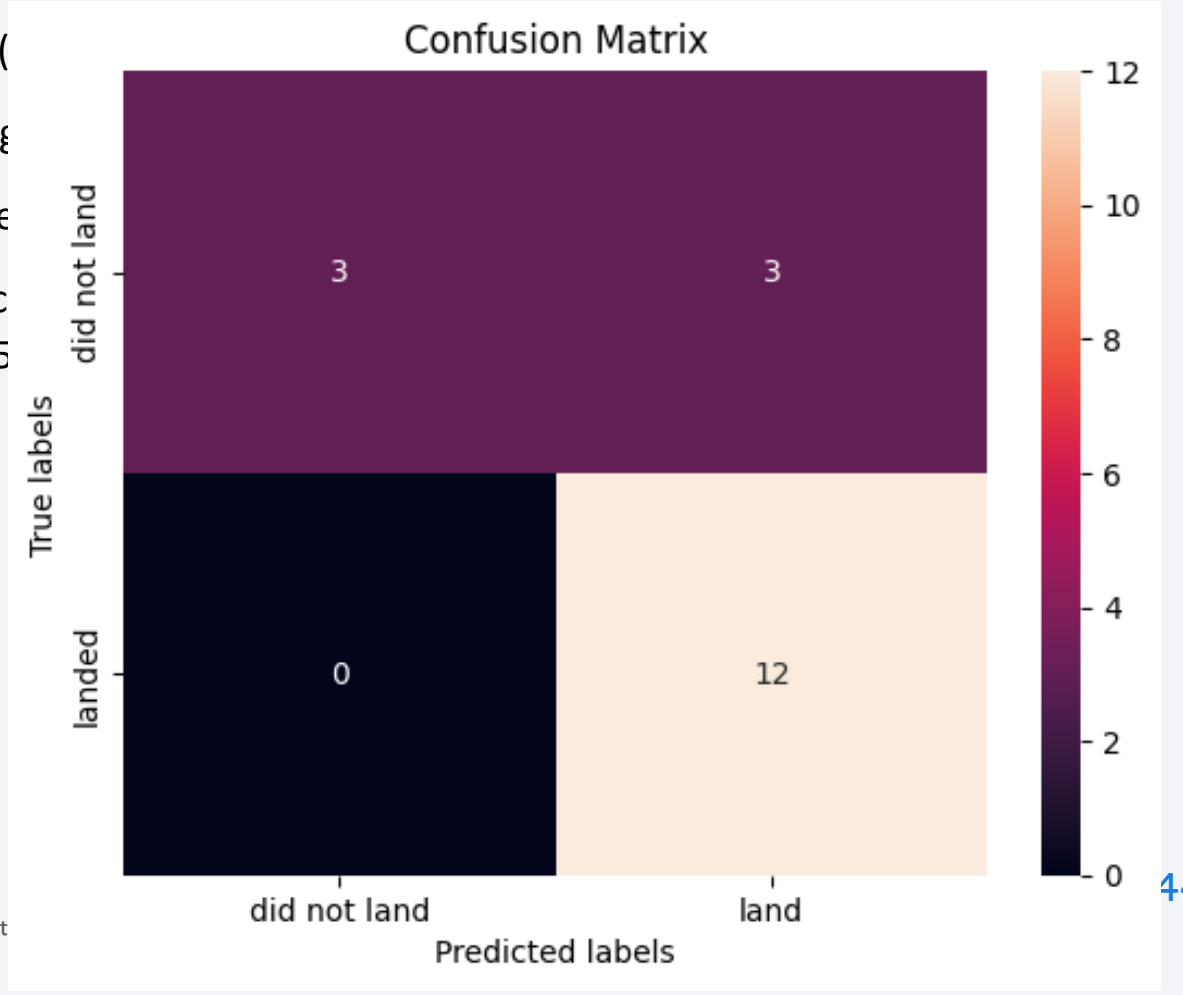
Predictive Analysis – Model Accuracy Comparison

- Test accuracies (4 classification models): Logistic Regr
  = 0.8333, SVM = 0.8333, KNN = 0.8333, Decision Tree
  0.7778.
- Cross-validation accuracies (GridSearchCV, 10-fold): S
  0.8482, KNN 0.8482, LR 0.8464, Decision Tree 0.8768.
- Selected best model: Logistic Regression (test accurac
  83.3%) due to robust performance on small test set (r
- Performed in: Modul-04_SpaceX_Machine Learning
  Prediction_Part_5 (1).ipynb.



Predictive Analysis – Confusion Matrix (Best Model: Logistic Regression)

- Best-performing model on test set: Logistic Regression (
- Confusion matrix: correctly classified majority of landing
- Interpretation: model generalizes well despite limited te
- GitHub: https://github.com/KhalidMohd99/Data-Scienc 04_SpaceX_Machine%20Learning%20Prediction_Part_5



Conclusions

- 1. Data collection via SpaceX API + Wikipedia scraping yielded clean dataset (90 Falcon 9 missions).
- 2. Overall first-stage landing success rate: 66.7%; success improved significantly over 2010–2020.
- 3. Launch site matters: KSC LC-39A and VAFB SLC-4E show higher success ratios than CCAFS SLC-40.
- 4. Orbit type influences success: VLEO/SSO/HEO/GEO/ES-L1 show high success; GTO is most challenging.
- 5. Payload mass correlates with orbit & success; very heavy payloads (>8–10t) can be viable but context-dependent.
- 6. Interactive mapping confirms CCAFS proximity to coastline (~0.58 km) supports recovery strategy.
- 7. Dashboard reveals site-specific patterns and payload bands driving outcomes.
- 8. Machine learning: Logistic Regression achieved best test

Appendix – Supplementary Assets

- Additional materials included:
- • All executed Jupyter notebooks (01–06 + ML) with preserved outputs.
- • Python app: spacex_dash_app.py (runnable dashboard).
- • Datasets (local CSVs): dataset_part_1/2/3.csv, spacex_web_scraped.csv, Spacex.csv, spacex_launch_geo.csv.
- • Database: 04-my_data1.db (10 SQL queries executed).
- • Assets: 7 EDA charts, 5 ML plots (4 confusion matrices + accuracy comparison), 5 Dash screenshots, 3 Folium map screenshots, 10 SQL result screenshots, flowcharts.
- • Scripts: fix_notebooks.py, extract_assets.py, build_dash_lab.py, gen_maps.py, screenshot_assets.py, build_ml_assets.py, build_flowcharts.py, deck_builder.py.
- • Structure Of A Report.pdf (reference).
- Complete repository: https://github.com/KhalidMohd99/Data-Science-Capstone-Project



Thank you!

Prepared by: Khalid Mohammed

Date: Oct 16, 2026

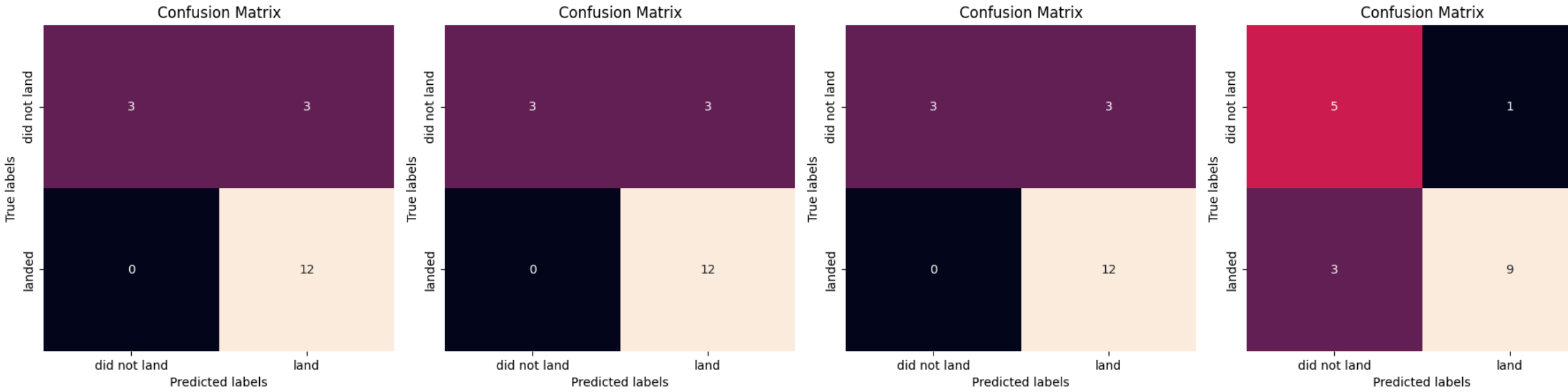
GitHub Repository: https://github.com/KhalidMohd99/Data-Science-Capstone-Project

This presentation completes the IBM Data Science Capstone Project.

All analyses, code, and assets are available in the repository for full reproducibility.

Roadmap: Planning → Data Collection → Wrangling → EDA → Interactive Analytics → Modeling → Dashboard → Presentation.
Tools: Python, Jupyter, pandas, numpy, matplotlib, seaborn, scikit-learn, folium, Plotly Dash, SQLite (ipython-sql), BeautifulSoup, requests.
Environment: Local venv, reproducible (notebooks + .py files), exports saved locally.
Creativity beyond template: consistent footer with GitHub URL, KPI framing in Executive Summary, flowcharts, well-organized asset folders (assets/eda,ml,da
Reproducibility: all notebooks executed, outputs preserved, CSVs included, scripts for screenshots & map generation.
GitHub: clean structure, all completed labs & Python files included.
Repository: https://github.com/KhalidMohd99/Data-Science-Capstone-Project

Test set: 18 instances (stratified 80/20).
Accuracies (test): LR 0.8333, SVM 0.8333, KNN 0.8333, DT 0.7778.
CV (GridSearchCV 10-fold): DT 0.8768 (best CV) but lower test (overfit tendency); LR/SVM/KNN stable ~0.846–0.848 CV, 0.8333 test.
Selected: Logistic Regression — simplest, stable, most generalizable on small test set.
Best hyperparameters (from tuning): LR {'C': 0.01, 'penalty': 'l2', 'solver': 'lbfgs'}; SVM {'C': 1.0, 'gamma': 0.0316, 'kernel': 'sigmoid'}; KNN {'algorithm': 'auto', '



- Learning curve is real: early flights (2010–2015) had lower success (~0–33%); post-2016 rose to ~83–90% (engineering maturity + reusability).
- Site reliability: KSC LC-39A (77.3%) and VAFB SLC-4E (76.9%) outperform CCAFS SLC-40 (60.0%) — site choice matters.
- Orbit challenge: GTO (27 launches, 51.9%) is hardest; VLEO/SSO/ES-L1/HEO show >85% success.
- Payload–success nuance: 8–12t payload bands achieve highest success (~87%); 4–8t bands lowest in the observed data.
- Recovery strategy is data-informed: CCAFS LC-40 sits ~0.58 km from coastline, enabling ocean recovery; geographic constraints drive site decisions.
- Ensemble parity: Logistic Regression/SVM/KNN tie at 83.3% test accuracy — with n=18 test samples, model simplicity (LR) is the prudent choice.
- Temporal trend is actionable: predicting near-future outcomes benefits from models trained on recent, successful operational profiles.
- Dashboard enables mission planners to explore trade-offs (payload range vs site) interactively.